Unit 4: Database Design & Development

Authorised Assignment Brief

Student Name/ID Number	
Unit Number and Title	**Unit 4: Database Design & Development**
Academic Year	2025
Unit Tutor	
Assignment Title	**Developing A Relational Database System**
Issue Date	3rd Feb, 2025
Submission Date	12th May, 2025

Submission Format

The assignment submission is in the form of:

- a **relational database design document** to design a relational database system for a substantial problem. The recommended word limit is 1,000–1,500 words, although you will not be penalised for exceeding the total word limit
- a **database development document** to document the development and testing of a fully-functional relational database system, based on an existing system design. The recommended word limit is 1,000–1,500 words, although you will not be penalised for exceeding the total word limit
- **user and technical documentation** for a fully-functional system, describing how the system works. The recommended word limit is 1,000–1,500 words, although you will not be penalised for exceeding the total word limit.

You are required to make use of headings, paragraphs and sub-sections as appropriate, and all work must be supported with research and referenced using an appropriate referencing system such as Harvard. You will also need to provide a bibliography using an appropriate referencing system such as Harvard. Inaccurate use of referencing may lead to issues of plagiarism if not applied correctly.

Unit Learning Outcomes
LO1 Use an appropriate design tool to design a relational database system for a substantial problem **LO2** Develop a fully-functional relational database system, based on an existing system design **LO3** Test the system against user and system requirements **LO4** Produce technical and user documentation.

Transferable skills and competences developed

Computing-related cognitive skills

- Demonstrate knowledge and understanding of essential facts, concepts, principles and theories relating to computing and computer applications
- Use such knowledge and understanding in the modelling and design of computer-based systems for the purposes of comprehension, communication, prediction and the understanding of trade-offs
- Recognise and analyse criteria and specifications appropriate to specific problems, and plan strategies for their solutions
- Critical evaluation and testing: analyse the extent to which a computer-based system meets the criteria defined for its current use and future development
- Methods and tools: deploy appropriate theory, practices and tools for the design, implementation and evaluation of computer-based systems

Computing-related practical skills

- The ability to specify, design and construct reliable, secure and usable computer-based systems
- The ability to evaluate systems in terms of quality attributes and possible trade-offs presented within the given problem
- The ability to deploy effectively the tools used for the construction and documentation of computer applications, with particular emphasis on understanding the whole process involved in the effective deployment of computers to solve practical problems
- The ability to critically evaluate and analyse complex problems, including those with incomplete information, and devise appropriate solutions, within the constraints of a budget.

Generic skills for employability

- Intellectual skills: critical thinking; making a case; numeracy and literacy
- Self-management: self-awareness and reflection; goal setting and action planning Independence and adaptability; acting on initiative; innovation and creativity
- Contextual awareness, e.g. the ability to understand and meet the needs of individuals, business and the community, and to understand how workplaces and organisations are governed.

Vocational scenario

You have been employed as a Junior Database Developer by Apex Technology Systems (ATS) Limited*. ATS provides a range of bespoke desktop-, web- and cloud-based design, consultancy and development services to both small- and medium-sized clients from across a diverse range of business sectors.

ATS has received an enquiry from a new client and your project manager has decided that this would be ideal for you to handle by yourself as your first solo project.

The client is Manzaneque Limited* (ML), a large real-estate company that specialises in residential, commercial and industrial properties. It has a large IT department, which supports a range of business operations including development of raw land, property management, brokerages, lending and other professional services such as lawyers, interior designers and construction workers. Owing to further expansion, ML has decided to establish a dedicated IT Helpdesk Team to handle hardware and software problems concerning the in-house IT systems.

ML have approached the Chief Executive Officer (CEO) of ATS with their requirements and the CEO has agreed, along with the Chief Technology Officer (CTO) and your line manager, that you would be the ideal candidate to design and build the new system. Your role will include designing, developing and implementing database systems based on ML requirements. You will also be responsible for optimising the database system for performance efficiency, as well as testing and troubleshooting and performing bug fixes.

The requirements of the system are as follows:

When someone in the company has a problem, they can contact the helpdesk. One of the five helpdesk operators will log the details of the job, and then allocate the job to one of several dedicated IT technicians. ML requires an Information System to log and track the helpdesk queries. This will enable analysts to see how the equipment is performing overall, and to assess the overall efficiency of the IT technicians in being able to solve problems.

You have been given the following information about the system requirements:

ML contains 15 offices around the country; each office has a name, address, contact phone and specialisation (land development, property management, brokerages, lending legal, interior designers and construction). Each office has between 10 and 20 employees, who have an Employee ID, Name, Contact Phone and Email, Employment Start Date and End Date. The system should record information about ex-employees also.

When a new call comes into the helpdesk, the names of the caller and helpdesk operator are logged, along with the time of the call, the serial number of the computer and, if relevant, the operating system and software being used.

The caller's name will be checked against a register of all personnel to retrieve the caller's ID number, job title and department.

Their equipment will also be checked against a register of equipment to find the equipment type and make. Their software will be checked to see if it is under a valid licence. Hardware will be checked to see if it is still under manufacturer warranty.

The equipment register contains the following:

- office hardware: PCs, laptops, printers, peripherals, interactive whiteboards
- network hardware: routers, switches, wireless access points, firewall, cables

- software: spreadsheets, word processors, presentation software, operating systems, time management, project management
- all relevant serial numbers, licence numbers, manufacturer, make and model details.

Every call is logged, and each problem is given a problem number, which is supplied to the caller so it can be quoted on any subsequent calls about the same problem. The date and time of the call are also logged.

The helpdesk operator will also record notes and descriptions of the problem. A reason for each call is always recorded even if it is just a note to indicate the urgency or distress of the caller, (e.g. in the case of a follow-up call).

When a problem is first reported, the helpdesk operator will also allocate the problem to an available IT technician who has a specialism in that area. Different IT technicians can have different specialisms e.g. Technician A can specialise in routers, laptops, printers and Office software, Technician B is a specialist in printers only.

Once a problem type has been identified, the system will display a list of only those IT technicians with the skills to solve that problem.

The system will also list how many problems the technician is currently addressing. This allows for load balancing, so if there is more than one technician for a given problem type, the one with the lightest workload can be assigned.

When a problem is eventually resolved, the helpdesk operator or the technician will log the date and time it is resolved, and record some indication of how the problem was resolved and the time taken (in hours) to resolve the problem.

The system will then need to provide the following reports to the management of ML, to see how well the technicians and equipment are functioning:

- a summary report of the total calls logged in a given month, showing the number of open and closed cases, as well as the total time spent on each closed case.
- a summary of all the technicians' job sheets in a given month, showing the technician's name, number of jobs worked, number of open and closed cases and total time spent
- a summary of the total jobs logged by each office, showing the office name, number of hardware faults, number of software faults, time spent and number of different IT technicians assigned
- a summary of the total jobs assigned to each item in the equipment register, to identify any items that need replacing or repaired under warranty/licence
- a report for a selected technician to show detailed information on the jobs they have worked on each month, including the office, job number and equipment details; the report should also show the total number of open and closed jobs and how long they have spent in total and on average on a job
- a summary report for a selected office to show detailed information on the jobs they have logged each month, showing the total number of open and closed jobs as well as summary counts of the number of jobs per register item to identify items of equipment that are regularly failing and will need replacing.

The system initially does not need to model all of this data but it should contain the basic relational database design to store a core level of pilot data and new data can be added at a later date.

The system should implement some form of application security measures to ensure that:

- helpdesk operators can log jobs but cannot open any reports
- office managers cannot log jobs but can view all the reports.

The system should use either inbuilt security functions in the software of your choosing or it can be implemented using your own security functionality.

The system should also implement some form of database maintenance routine, that includes:

- backing up the database
- restoring the database from the backups
- compacting the database.

The maintenance backup operations should run from a command on the main control page and should be accessible by everyone. Restoration and compaction of a database should be performed only by helpdesk operators.

Before beginning the implementation of the system, the boards of both ML and ATS are interested in your designs for a relational database system, including any user interface that you intend to produce.

Once you have implemented the database system based on your designs, the boards of both ML and ATS are keen that the system be thoroughly tested to make sure that it meets the ML client requirements.

Finally, ML would like to examine any user and technical documentation that you produce so that when the database application is implemented by the ML helpdesk team, they have a thorough understanding of not only how the system works but also how it was designed and implemented so that they can identify any issues or further requirements. The user documentation should clearly show how to use the system, menu operations and system outputs, as well as other functions. Technical documentation should clearly show the movement of data through the system.

*ATS and ML are not real companies. Real or fictitious companies can be chosen to meet local assessment needs.

Assignment activity and guidance

Relational Database Design Document

You will create a design document for the boards of ML and ATS in which you will design a relational database system for the given problem. You will:

- design a relational database system using appropriate design tools and techniques
- include at least four interrelated tables
- provide clear statements of user requirements
- provide clear statements of system requirements.

The comprehensive design for a fully-functional system will include:

- interface designs
- output designs
- data validation designs
- data normalisation designs.

You should then go on to evaluate the effectiveness of the design in relation to:

- user requirements
- system requirements.

Database Development Document

You will create a database development document for the boards of ML and ATS in which you will develop a fully-functional relational database system, based on an existing system design. Your document will show:

- the development of the database system
- evidence of the following:
  - user interface
  - output
  - data validations
  - querying across multiple tables
- evidence of the implementation of a query language into the relational database system
- implementation of a fully-functional database, which includes:
  - system security
  - database maintenance
- an assessment of whether meaningful data has been extracted through the use of query tools to produce appropriate management information
- testing of the system against:
  - user requirements
  - system requirements
- an assessment of the effectiveness of the testing
- an explanation of the choice of test data used.

Finally, your development document will evaluate the effectiveness of the database solution in relation to user and system requirements and suggest improvements.

User and Technical Documentation

You will create user and technical documentation for the boards of ML and ATS. The documentation of the fully functional system will describe how the system works and should include:

- data flow diagrams
- flowcharts.

At the end of the documentation, you will evaluate the database in terms of improvements needed to ensure the continued effectiveness of the system.

Recommended resources

Please note this is not a definitive list of resources but it will help you begin your research by acting as a starting point of reference.

Websites

https://blog.airtable.com (2023) *How to design an effective relational database* [online].Available at: https://blog.airtable.com/how-to-design-an-effective-relational-database/ [Accessed 10 June 2024]

https://online.visual-paradigm.com (n.d.) *Data Modeling: Conceptual vs Logical vs Physical Data Model* [online]. Available at: https://online.visual-paradigm.com/knowledge/visual-modeling/conceptual-vs-logical-vs-physical-data-model [Accessed 10 June 2024]

https://support.microsoft.com/en-GB (2022) Database design basics [online]. Available at: https://support.microsoft.com/en-us/office/database-design-basics-eb2159cf-1e30-401a-8084-bd4f9c9ca1f5 [Accessed 10 June 2024]

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https://www.oracle.com (n.d.) *What is a relational database (RDBMS)?* [online]. Available at: https://www.oracle.com/uk/database/what-is-a-relational-database/ [Accessed 10 June 2024]

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Journals and articles

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HN Global

Reading lists. Available at:

https://hnglobal.highernationals.com/learning-zone/reading-lists

Student Resource Library. Available at:

https://hnglobal.highernationals.com/subjects/resource-libraries

Learning Outcomes and Assessment Criteria

<table>
<tr><th>Pass</th><th>Merit</th><th>Distinction</th></tr>
<tr><td colspan="2">LO1 Use an appropriate design tool to design a relational database system for a substantial problem</td><td rowspan="2">D1 Evaluate the effectiveness of the design in relation to user and system requirements.</td></tr>
<tr><td>P1 Design a relational database system using appropriate design tools and techniques, containing at least four interrelated tables, with clear statements of user and system requirements.</td><td>M1 Produce a comprehensive design for a fully-functional system, which includes interface and output designs, data validations and data normalisation.</td></tr>
<tr><td colspan="2">LO2 Develop a fully-functional relational database system, based on an existing system design</td><td rowspan="5">LO2 and LO3
D2 Evaluate the effectiveness of the database solution in relation to user and system requirements and suggest improvements.</td></tr>
<tr><td>P2 Develop the database system with evidence of user interface, output and data validations, and querying across multiple tables.
P3 Implement a query language into the relational database system.</td><td>M2 Implement a fully-functional database system, which includes system security and database maintenance.
M3 Assess whether meaningful data has been extracted through the use of query tools to produce appropriate management information.</td></tr>
<tr><td colspan="2">LO3 Test the system against user and system requirements</td></tr>
<tr><td>P4 Test the system against user and system requirements.</td><td>M4 Assess the effectiveness of the testing, including an explanation of the choice of test data used.</td></tr>
<tr><td colspan="2">LO4 Produce technical and user documentation.</td><td rowspan="2">D3 Evaluate the database in terms of improvements needed to ensure the continued effectiveness of the system.</td></tr>
<tr><td>P5 Produce technical and user documentation.</td><td>M5 Produce technical and user documentation for a fully-functional system, including data flow diagrams and flowcharts, describing how the system works.</td></tr>
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